

LESSON 14
BUSINESS MATHEMATICS

Overview:

This lesson will introduce basic concepts in business Mathematics such as simple interest, compound interest, and annuity. It will provide problems based in real-life scenarios.

Learning Outcomes:

At the end of this lesson, the students can:

- 1. compute interest of an investment; and
- 2. solve problems on simple interest, and compound interest.

Materials Needed:

- Module (Printed/Softcopy)
- Internet Connection
- Smartphone/Laptop, Google Classroom, FB Group Chat and YouTube videos
- Calculator

Duration: 2 hours

Learning Contents:

14.1 SIMPLE INTEREST

14.1.1 Principal

The amount borrowed or deposited for a period is called **principal (P)**.

14.1.2 Interest

An **interest (I)** is an extra amount charged on principal for a certain period of **time (t)** at a uniform **rate (r)**.

$$I = \frac{P \times r \times t}{100}$$

SIMPLE INTEREST

Where,

I – Interest

P – Principal

r – interest rate (in decimal form)

t -- time (in years)

EXAMPLE 14.1: How much interest is earned when an amount of ₱100,000 is invested with an annual interest rate of 8% for 5 years?

Solution:

Given: $P = 100,000.00$, $r = 8\%$, and $t = 5$

Find: $I = ?$:

$$I = P r t = 100,000 \times 0.08 \times 5 = ₱40,000.00$$

Hence, an interest of ₱40,000 will be earned from ₱100,000 principal amount in 5 years.

EXAMPLE 14.2: How much amount will the investor in problem 1 received in 5 years?

Solution:

Given: $P = 100,000.00$ and $I = 40,000.00$

Find: $A = ?$

When the principal amount is added to the interest, the sum is called the **total amount (A)**.

TOTAL AMOUNT	
(i) $A = P + I$,	Where,
$= P + P r t$	I – Interest
$= P(1 + r t)$	P – Principal
(ii) $A = P(1 + r t)$	r – interest rate (in decimal)
	t – time (in years)

Using the formula (i), $A = P + I = 100,000 + 40,000 = 140,000$

Therefore, the investor will receive an amount of ₱140,000 after 5 years.

EXAMPLE 14.3: A businessman invested ₱80,000 in a firm which will earn an annual interest of 12% for 2 years. But financial needs arise and he has to take them out in 9 months only. How much amount will he receive?

Solution:

Given: $P = 80,000$, $r = 12\%$, and $t = 9 \text{ mons.} = 9/12 = 0.75 \text{ year}$

Using the formula (ii), $A = P(1 + r t)$

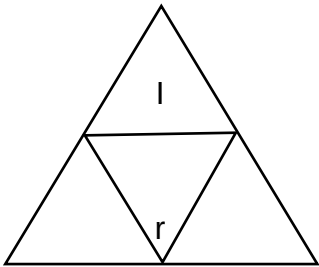
$$= 80,000 [1 + (0.12)(0.75)]$$

$$= 80,000 (1 + 0.09)$$

$$A = 87,200$$

Thus after 9 months, the business man will receive ₱87,200.

In the simple interest formula, one missing value can be found given the other values.



if P is missing	$P = I / r.t$
If r is missing	$r = I/Pt \times 100$

14.2. COMPOUND INTEREST

Compound interest is interest earned on an interest. In the compound interest, the rate of interest is taken from the accumulated amount (principal for the period plus the interest of the period or term).

It can be computed with the following formula.

$$A = P (1 + r/c)^{c.t}$$

Where, A – amount (future value)
 P – Principal (present value)
 r – rate (in decimal)
 t – number of years

Note: c = 1 (annually) c = 2 (semi - annually) c = 4 (quarterly)
 c = 12 (monthly) c = 365 (daily)

EXAMPLE 14.4-6: Suppose ₱10,000 is invested in account paying 6% compounded quarterly.

4. How much will be in the account at the end of 5 years?

Solution:

Given: P = 10,000 , r = 6% = 0.06, t = 5, and c = 4 (quarterly)

Find: A = ?

$$A = P (1 + r/c)^{c.t} = 10,000(1 + 0.06/4)^{4(5)} = 10,000(1 + 0.015)^{20}$$

$$= 10,000(1.3469) \text{ *to four decimal places}$$

$$A = 13,468.55$$

5. How much interest will it earn?

Solution:

Recall that the total amount is the sum of the principal amount and its interest, that is $A = P + I$. From this,

$$\begin{aligned} I &= A - P \quad \text{where: } A = 13,468.55 \text{ and } P = 10,000 \\ &= 13,468.55 - 10,000 \\ I &= 3,468.55 \end{aligned}$$

Thus, the interest will earn ₱3,468.55.

6. Compare the interest it will earn if it is paid at simple interest rate.

Solution:

Using the simple interest formula, we have

$$\begin{aligned} I &= Pr.t \quad \text{where: } P = 10,000, \quad r = 6\% = 0.06, \text{ and } t = 5 \\ &= 10,000 \times 0.06 \times 5 \\ I &= 3,000 \end{aligned}$$

Thus, comparing the interest that will be earned using the simple interest, compound interest will earn ₱468.55 more than the simple interest.

Learning Activity:

I. Find the missing values in the table. (Round answers to the nearest hundredths if needed).

<i>I</i> (in pesos)	<i>P</i> (in pesos)	<i>r</i>	<i>t</i> (in years)
(1)	8,000	4.5%	4 months
300	(2)	8%	5 years
980	4,800	6%	(3)
(4)	18,654	9.05%	18 months
6,800	(5)	11.5%	1.4 years

II. Complete the table below by finding the required values.

<i>A</i> (future value)	<i>P</i> (present value)	<i>r</i> (rate)	<i>c</i> (no. of compounding)	<i>t</i> (time)
(1)	₱800	8%	semi-annually	3 years

₱1,800,000	(2)	12%	quarterly	10 years
₱150,000	₱25,000	(3)	annually	3.5 years
₱175,000	₱54,500	6.25%	semi-annually	(4)
₱78,890	₱25,540	(5)	monthly	12.2 years

Learning Evaluation:

Solve each of the following problems.

- Grace invested her ₱150,000 to a money market which earns an annual simple interest of 8.5% for five years. If she is 45 years old now, how much will she receive after her retirement at 65?
- How long does it take for a ₱240,000 investment at 13% compounded quarterly to be worth more than ₱300,000 investment at 8% compounded quarterly?
- One investment pays 12% simple interest and one pays 9% compounded annually. Which investment is best to have a greater profit and by how much?
- If ₱15,000 is loaned for four months at 4.5% annual interest, how much interest is earned?
- How much interest will one pay for a credit card statement of ₱9,760 that is two months overdue if a 12.5% annual rate is charged?

REFERENCES:

Banigon Jr., R. et. Al. (2016). *Statistics and Probability, Educational Resources Corporation*. 11 – B Sunrise Drive Barangay Bagong Lipunan ng Crame, Cubao Quezon City.

Banigon Jr., R. et. Al. (2016). *General Mathematics*. Educational Resources Corporation. 11 – B Sunrise Drive Barangay Bagong Lipunan ng Crame, Cubao Quezon City.

Aufman, R. et. Al. (2018), *Mathematics in the Modern World*, Rex Book Store Inc., Sampaloc, Manila, Philippines.

R. David Gustafson, et al., *Beginning Algebra*, 2012.

Commission on Higher Education. (2016). *General Mathematics [Teaching Guide]*, C.P. Garcia Ave., Diliman, Quezon City.
<https://www.academia.edu/29721020/>